IOT Based Automated Poultry Farm for Layer Chicken

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ABSTRACT—

The poultry industry plays a major role in the global food supply chain. It is providing a sustainable source of protein. According to data from the Food and Agriculture Organization of the United Nations (FAO), in 2020, chicken meat production reached a record high of approximately 102 million metric tons. It is approximately over 36% of global meat production. Most of the poultry farms in India are physically operational. As a result, traditional poultry farming methods face many challenges. The main challenges are labor-intensive processes, environmental impact, disease management etc . To solve these issues, automated poultry farms offer a viable option that optimizes chicken production operations. In this century Internet of Things (IoT) technology is a very efficient way to solve the limitations. In order to increase production, efficiency, and sustainability in chicken farming, this research study examines IoT applications. In addition to discussing earlier studies and research on IoT-based poultry farming systems. This paper includes their advantages and disadvantages. This paper provides an overview of the current state of IoT in chicken farming. The study's design and implementation of an IoT-based chicken farming system. It consists of sensors, wireless communication networks, and cloud-based data management systems, which are also covered in the report.

***Keyword*- IOT, Sensor, Ardino, Cloud**

1. INTRODUCTION

The poultry farming sector makes a considerable economic contribution to many nations. For the purposes of producing eggs, meat, and feathers. With new technologies and agricultural methods emerging to increase productivity, animal care, and financial success, there have been considerable developments in poultry farming. This essay will examine numerous facets of chicken farming, including its history, diverse farming systems, production

methods, difficulties, and opportunities for the future. Cloud-based data management systems, is also covered in the report

Layer chickens are known for their egg-laying capabilities. It can produce up to 300 eggs per year on average, which is equivalent to almost one egg per day. However, ensuring healthy chicken growth requires regular monitoring of various factors. The main and critical ones are temperature, air quality, humidity, ventilation, and lighting.

The automation of chicken monitoring processes can significantly contribute to the growth of the poultry trade. The management of temperature, air quality, ventilation, and other factors can be labor-intensive, and the proposed project aims to automate these processes using the Internet of Things (IoT). The system will monitor and control everything to ensure healthy chicken growth and quality egg production. The main focus of the project is on the expansion of layer chickens. Overall, this project seeks to leverage technology to optimize poultry farming processes and improve the productivity and efficiency of layer chicken production . while also ensuring the health and quality of poultry products through automated monitoring and management of critical factors.

2. DIFFERENT FARMING SYSTEM

In chicken farming, there are several agricultural systems employed. Every system has its own pros and cons. Free-range, cage, and barn systems are the most popular types.

Free-Range System: In a free-range system the birds are given unrestricted access to food, water, and shelter when they are outdoors. For this method it takes large space .Given that it gives birds more freedom to fly around and engage in natural activities, this method is frequently regarded as being more moral. But it also has some disadvantages. It has more susceptibility to disease and predators. It has higher production costs and increased exposure to both.

Cage System: In this system, birds are kept in cages with little room for movement, frequently in large-scale operations. This system has drawn criticism because of purported problems with animal welfare. On the bright side this method is well known for its great production and low labor costs.

Barn System: Keeping birds indoors in sizable barns with access to food, water, and environmental controls is known as the "barn system." This system provides a balance between output, animal welfare, and cost effectiveness. But this system needs a substantial investment in technology and infrastructure.

3. CHALLENGES IN POULTRY FARMING

Many difficulties face the poultry industry, including disease outbreaks, issues with food safety, and environmental effects.

Disease Outbreaks: Disease outbreaks can have a huge negative impact on the

economy and public health if they affect poultry production. Many nations have implemented trade restrictions and conducted large flockings of chickens in response to avian influenza, Newcastle disease, and other outbreaks of poultry diseases.

Food Safety Concerns: Worries about Food Safety Products made from poultry are frequently linked to foodborne illnesses, including salmonella and cyanobacteria. Antimicrobial resistance has been connected to procedures used in poultry husbandry, such as the use of antibiotics and other veterinary medications.

4. RELATED WORKS

In this research, to observe and manage the atmospheric conditions of the poultry farm we are using a wireless sensor network technology (WSN).. An electronic network technology will replace a major part of the farmer's work. This computerized system improves the quality of egg production. It is a combination of hardware and open source code for the management of the domesticated farm. It also includes dampness, temperature, air quality and light intensity.[1] The main focus of the system is to produce a cost efficient hardware and an open supply code and a setup like IOT. It will save time and it will reduce the reliance of manpower and ameliorate a fit environment. This will in turn increase poultry production, mainly eggs. The main stream of this analysis paper is to monitor and to administer the poultry farm.[2] By employing WSN technology to achieve a precise action. A system is being utilized by the user to provide a healthy feed for chicken and to control the atmospheric condition in the farm even when the user is not in the farm . To maintain a healthy ambiance in the poultry farm, environmental factors such as temperature, humidity, air quality have to be continuously monitored. The key focus of this paper is to automate the poultry farm with exploitation of wireless sensor networks and mobile communication systems By exploitation the quality of egg production is revamped.[3]

To make an automatic environment controlled poultry management system, is the main target of the system. System does not only monitor the poultry house, it additionally regulates these parameters properly . The system's approach is to manage the exploitation remotely through hand-held mobile devices. The paper focuses on producing the required atmospheric conditions and regulates the performance of the farm. The right methodology of dominant poultry barn is the oxygenating system and it additionally embraces a critical issue which is atmospheric temperature and atmospheric wetness .Temperature and humidity is a crucial factor for chicken health . This system does not solely scale back the production cost, it also improves the health of the chickens. The paper focuses on inculcating WSN and GPRS network to regulate and supervise the surrounding environmental factors such as temperature, humidity, ammonia gas etc. within the poultry farm. Food and water level is furthermore supervised and surveilled the exploitation of machine-controlled systems . It also helps the farmers to watch and remotely acquire the entire system. [4]One of the most important keys is to monitor the food intake by the chicken and the water level, on a time to time basis, in order to minimize wastages. The information being collected from the poultry's setting through a wireless sensing element is combined with the sensors. By this process atmospheric climate in the poultry is being monitored and it is well approachable to the user . This technique keeps track of the system and collects useful information from the poultry farm to the owner wirelessly and the system is giving live updates of a farm anytime .[5] Thus this technique is ready to congregate the information and play it mechanically and aids to regulate the surrounding temperature in the domesticated barn.

Indian poultry is one of the major contributors to the National food security. It depends on obstacles of the excess heat being generated in summer and cold waves during winter. If the temperature is extreme during the daylight or cold at night it will affect the food quality and the health of a chicken[6]. The paper mainly focuses on scanning the properties such as economy, chillness and warmth in poultry farms by exploitation of the earth-air device. The temperature is

additionally maintained by the exploitation of the wireless sensors. By utilizing this system, the quality and amount of eggs laid by the chicken is improved to maintain human health[7]. The production and economy of the poultry farm is achieved with the help of wireless sensing element nodes which is highly helpful for early detection of the well-being of the chickens. The development of the wireless sensor is focused by the analysis paper[8]. We are using small temperature sensor technology .It has better sensitivity and low power consumption .By this a sensible monitoring is achieved through minimum functions which eventually is the main focus of this paper.

5. EXISTING SYSTEM

The majority of the domestic barns in India utilize customary meat and egg cultivating techniques. The everyday tasks like monitoring temperature, moisture, food intake, air quality illuminating, oxygenating and warming are manually observed and administered by a farmer or labor . It is hard to monitor all the facts physically and it's impossible to take accurate measurements of all facts ,the error rate is high. It also compromises the hygienic fact. During dusk, if the administrator neglected to inspect or change the temperature, it can influence the well-being of the chicken. Any alteration in the moistness, it might influence the development of the chicken and its increase the chance of multiple disease .If the temperature and humidity suddenly goes up it will automatically trigger the exhaust fan and it will cool it down to the desire parameter .After the number hit the desire parameter the fans will automatically shut down . The domesticated bird dwelling place continuously demands great air quality in order to improve the well-being of the chicken. As of now, there could be no appropriate framework for checking the Ammonia gas in the outbuilding and on the off chance that they distinguish any overabundance ammonia gas, administrators switch on the fumes fan physically what's more, that will eliminate the gas to the outside of the outbuilding[9]. There will be lux sensors that can detect daylight if the sun goes down the lights bulbs will be on automatically. The farmer can also control all the system from his mobile devices because all the components of the system are connected with IOT devices.

6. PROPOSED SYSTEM

The aim of this project is to provide healthy chickens that produce quality eggs and to boost the production of eggs in India by monitoring, controlling ,controlling and stabilize any kind of fluctuation in environmental factors that can be harmful for chicken such as air quality, humidity, temperature, ventilation, lighting automatically and cultivating poultry feed through the Internet of Things (IoT).

A. Hardware requirements

- Arduino MEGA
- Power supply
- Temperature sensor
- GAS sensor
- Relay board
- LCD display
- WIFI module
- LED
- Weight sensors

B. Software requirements

- Arduino IDE
- Web Technologies
- Databases
- Google Firebase

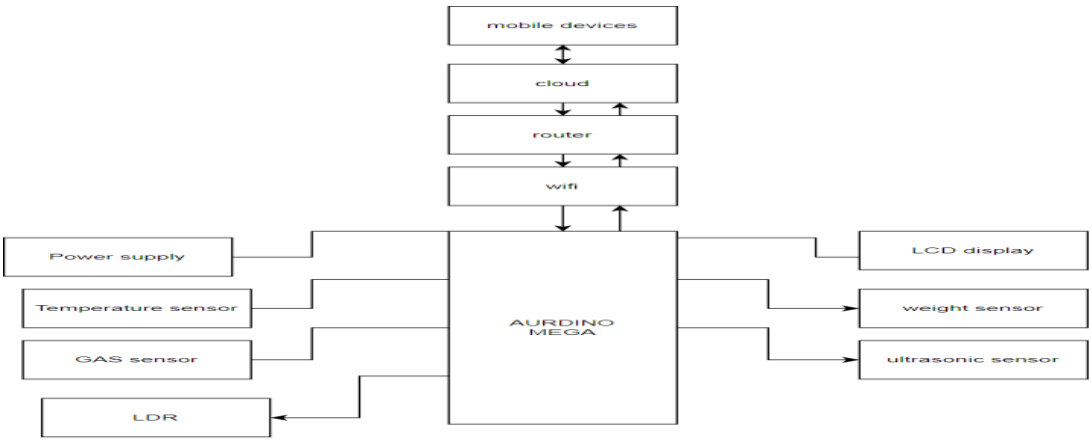

Fig. 1: Block Diagram

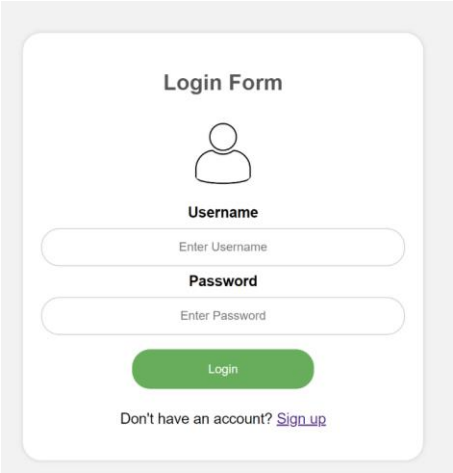

Fig. 2: Login and LogoutForm

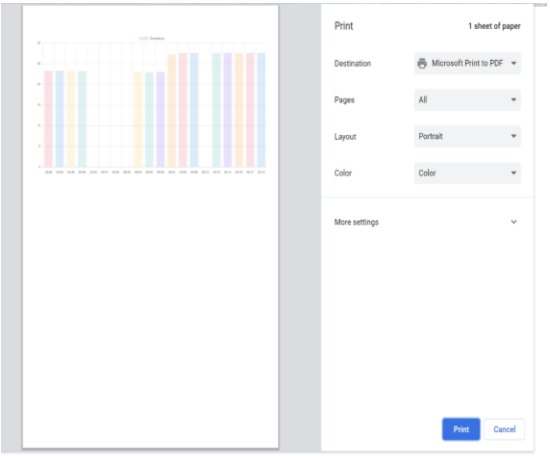

Fig. 4: Print The Data

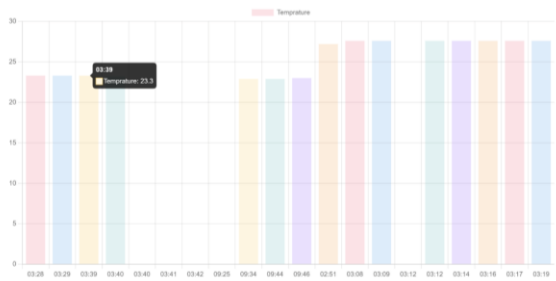

Fig. 3: Graph of Every Sensor Data

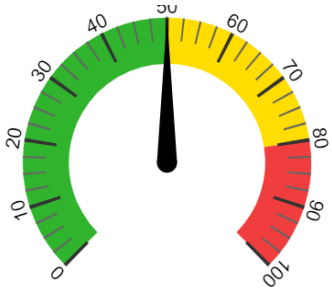

Fig. 5: Gauge of Every Sensor Data

7. RESULT

In the result it is shown that, in case of high or low temperature, the temperature sensor will monitor the change and give the indication. Then, in case any harmful gas such as Ammonia is formed in the farm, it can be monitored by the help of the gas sensor and can be cleared by exhaust fan ventilation. The temperature, humidity and air quality is uploaded in the cloud by the help of the wi-fi and displayed in the website. For the waste management system there is an ultrasonic sensor and weight sensor.

8. CONCLUSION

The environment of the farm is very important because it can affect the health of the chicken. If the atmosphere is not well monitored, the problem can turn into a medical issue for the chickens. It is quite hard to monitor everything physically and especially at night times. Our model displays the utilization of IOT technology for layer chicken farming to monitor the farm and control the environment of the farm. The model sends data to the cloud and displays it in the website for easy monitoring and automating the hitter, cooler, ventilator fan and west management system. So, the model is very effective to reduce manpower and save time also. To manage the increasing demand of chicken and eggs in the world, farm owners need healthy chickens and our model is highly effective for monitoring the good health of the chickens.

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